



CCS3102 – Mobile Computing
Faculty of Computing
University Of Sri Jayewardenepura

MediGuide

**Disease Prediction and Physician Recommendation Mobile
Application**

Project Proposal

Submitted by

Group No 08

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1. Problem Statement

Healthcare accessibility and early disease identification remain significant challenges faced by individuals and communities. Many people struggle to understand the seriousness of symptoms they experience and often delay seeking medical attention due to uncertainty, lack of knowledge, or difficulty identifying the appropriate medical specialist.

Patients frequently search online for symptoms, but such information sources are often unreliable, confusing, or generalized. In addition, finding suitable doctors based on specialization, availability, and location can be time-consuming. This situation may lead to delayed treatment, incorrect self-diagnosis, increased healthcare costs, and unnecessary pressure on healthcare providers.

The problem particularly affects:

- Individuals with limited healthcare knowledge
- Patients in rural or underserved areas
- People requiring quick guidance before consulting a physician
- Users seeking suitable doctors based on symptoms

Current healthcare applications generally provide appointment booking or health information separately, but there is limited integration between symptom analysis and physician recommendation. Users still need to manually identify diseases and search for doctors independently.

Therefore, there is a need for a mobile-based solution that assists users by analyzing symptoms, predicting possible diseases, and recommending suitable healthcare professionals in one integrated platform.

2. Objectives

The primary goal of MediGuide is to provide users with an intelligent healthcare assistant through a mobile application. Specific objectives include the followings.

Main Objectives

- Develop a mobile healthcare application compatible with Android devices.
- Provide symptom-based disease prediction functionality.
- Recommend doctors based on predicted diseases and specialties.
- Enable appointment booking with healthcare professionals.
- Improve healthcare accessibility and decision-making.

Technical Objectives

- Integrate AI-powered healthcare assistance features.
- Implement secure user authentication and profile management.
- Provide location-based doctor recommendations.
- Store and manage healthcare information using cloud databases.
- Design an intuitive and responsive user interface.

3. System Outline

Overview

MediGuide is a healthcare mobile application designed to assist users in identifying possible diseases and recommending suitable doctors based on symptoms entered by users.

The application combines symptom analysis, disease prediction, physician recommendation, appointment management, and healthcare record tracking within a single platform.

Users can register and create profiles containing health information such as age, allergies, blood group, and medical history. They can then enter symptoms through an interactive symptom checker.

The system processes these symptoms using a prediction engine and generates possible diseases along with confidence scores. Based on prediction results, the system recommends nearby healthcare specialists according to specialization, availability, ratings, and user location.

Users may further schedule appointments and receive notifications regarding appointments and health updates.

Functional Requirements of the System

User Authentication

- User registration
- Login/logout
- Password reset using email OTP
- JWT session management

Profile Management

- Edit personal information
- Manage allergies and medical conditions
- Store health records

Symptom Checker

- Searchable symptom list
- Symptom autocomplete
- Multi-selection capability
- Severity and duration input
- Symptom history records

Disease Prediction

- Symptom-based disease analysis
- Confidence score generation
- Disease descriptions
- Emergency condition detection

Doctor Recommendation

- Specialist matching
- Doctor profiles
- Rating display
- Location-based filtering

Appointment Management

- Schedule appointments
- Booking history

- Appointment status updates

Notifications

- Appointment reminders
- Health tips
- Booking alerts

Non-Functional Requirements

The proposed system should demonstrate:

- Responsive user interface design
- Good user experience
- Fast performance
- Code maintainability
- Reusable components
- Scalability
- Security and privacy protection
- High availability

4. Tools and Technologies

Layer	Technology
Mobile Frontend	React Native (Expo SDK 54) + TypeScript
Backend API	Node.js + Express.js + TypeScript
Database /Backend services	Supabase (PostgreSQL + Authentication + Cloud Storage)
Authentication	JWT + Supabase Auth
State Management	React Context API + AsyncStorage
Navigation	React Navigation v6
HTTP Requests	Axios
Styling	React Native StyleSheet
Push Notifications	Expo Notifications
Maps and Location	React Native Maps + Expo Location
Image Handling	Expo Image Picker
AI Integration	Gemini API
Machine Learning	SVM prediction Model

Version Control	GitHub
Development Environment	VS Code
API Testing	Postman

5. Initial Plan / Task Allocation

Task	Responsible Member
Requirement gathering and proposal preparation	R.M.L.G.Rathnadoowa
UI wireframes and application design	T.D.J.Amaranath
Authentication and profile management	H.H.D.Nagahawatta
Symptom checker module	R.M.L.G.Rathnadoowa
Disease prediction engine	H.H.D.Nagahawatta
Doctor recommendation functionality	T.D.J.Amaranath
Appointment management	R.M.L.G.Rathnadoowa
Backend API integration	H.H.D.Nagahawatta
Database implementation	T.D.J.Amaranath
Testing and debugging	All
Documentation and presentation	All
